AUTUMN END SEMESTER EXAMINATION-2018

3rd Semester B.Tech & B.Tech Dual Degree

WEB TECHNOLOGY

IT-2003 / IT_2003

[For 2018(L.E.) & 2017 Admitted Batches]

Time: 3 Hours Full Marks: 50

Answer any SIX questions including question No.1 which is compulsory.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

1. Answer all the questions. [1 × 10]

(a) Find the output/errors of the following program?

```
class A {
        int x=10;
        void  f1(){ System.out.printlln("Hello");}
        }
class B extends A{
        int x=20;
void  f1(){ System.out.printlln("Bye");}
}
class Demo{
public static void main(String z[]){
A a= new B();
System.out.println(a.x);
a.f1();
}
}
```


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(b) Find the output/errors of the following program?

```
classTest
{
  static int a;
  static
  {
    a = 4;
    System.out.println ("inside static block\n");
    System.out.println ("a = "+ a);
  }
  Test()
  {
    System.out.println ("\ninside constructorn");
    a = 10;
  }
  public static void func()
  {
    a = a + 1;
    System.out.println ("a = "+ a);
  }
  public static void main(String[] args)
  {
    Test obj = newTest();
    obj.func();
  }
}
```

(c) Write down the html code for the links which are used to traverse within the single page

BOTTOM

Home page Home page Home page

Home page Home page Home page

TOP

(d) Why can abstract class not be instantiated?

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(e) Explain the percentage and decimal notation to specify the size of frames in HTML.

(f) What is garbage collection? How does one call it explicitly?

(g) Explain the use of toString() method with suitable example.

(h) What is the advantage of default access specifier over public access specifier in java?

(i) Explain definition list with example in HTML?

(j) Explain the types of instance variables defined in System class.

2. Write a program in HTML to create following table. [8]

Favorite and Least Favorite Things

		Bob	Alice
Favorite	Color	Blue	Purple
	Flavor	Banana	Chocolate
Least Favorite	Color	Yellow	Pink
	Flavor	Mint	Walnut

Write a program in HTML to create following employee registration form

Attributes	**Type**
FirstName	-Textbox
LastName	-Textbox
BirthPlace	-Textbox
Gender	- radio buttons
PrePosition	-Textbox
Marital Status	-Drop Down box with yes/no values
Date of Birth	-combo box with day, month & year
Social Security Number	-number
Nationality	-Drop Down box
Address	-text area
Mobile number	-number with max size 10
Submit	-button

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3. (a) Write a program to create a class SavingsAccount.Use a static variable annualInterestRate to store the annual interest rate for all account holders.Each object of the class contains a private instance variable savingsBalance indicatingthe amount the saver currently has on deposit.Provide method calculateMonthlyInterest to calculate the monthly interest by multiplying the savingsBalance by annualInterestRate divided by 12 and this interest should be added to savingsBalance. Provide a static method modifyInterestRate that sets the annualInterestRate to a new value.Write a program to test class SavingsAccount.Instantiate two savings Account objects, saver1 & saver2,with balances of $2000.00 and $3000.00 respectively. Set annualInterestRate to 4%,then calculate the monthly interest and print the new balances for both savers. Then set the annualInterestRate to 5%,calculate the next month's interest and print the new balances for both savers. [4]

(b) What is abstraction? How encapsulation, polymorphism and inheritance facilitates abstraction ? Explain with suitable example. [4]

4. (a) Implement the concept of abstract class for the following: [4]

Create a base class as conversion defined with data member amount and the abstract method MoneyConversion(). Then create two SubClasses named as RupeetoDollar and RupeetoEuro, then implement the abstract method Moneyconversion() in these classes. Finally create an object for the SubClasses and test it?

(b) Write a program to read 20 marks from keyboard and store them in an array. Define an user defined exception called WrongMarkException. Throw and catch this type of exception when marks are less than zero or marks are greater than 100. [4]

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5. (a) Write a program to create an interface LCM_GCD with methods findLcm(int x, int y) and findGcd(int x, int y). [4]

Create a class MathOperations which implements the interface and find lcm and gcd of two numbers.

(b) What is inner class? Explain it's characteristics. Also discuss it's application in event handling with suitable example. [4]

6. (a) Create a window application to calculate the Factorial and Square of a given number.Create a textfield to take the number as a user input for these operations.To perform the calculation, create two buttons namely FACTORIAL and SQUARE.Create another textfield to display the results. [4]

(b) Write a program to create a class student having studentId, Name, age, Branch, Section, University as data members. Now display three student's details using the ArrayList class. [4]

7. (a) Write a program that takes an input from a text file and replaces a '#' by a '*' and replaces every new line with '/' and stores the output to another file. [4]

(b) Write a program in java to explain following StringBuffer class methods. [4]

a) indexOf()

b) ensureCapacity()

c) delete()

d) append()

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8. Write short notes on *any two* of the following [4×2]

(a) for each statement

(b) Applet life cycle methods

(c) Style Sheets

(d) finally block

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